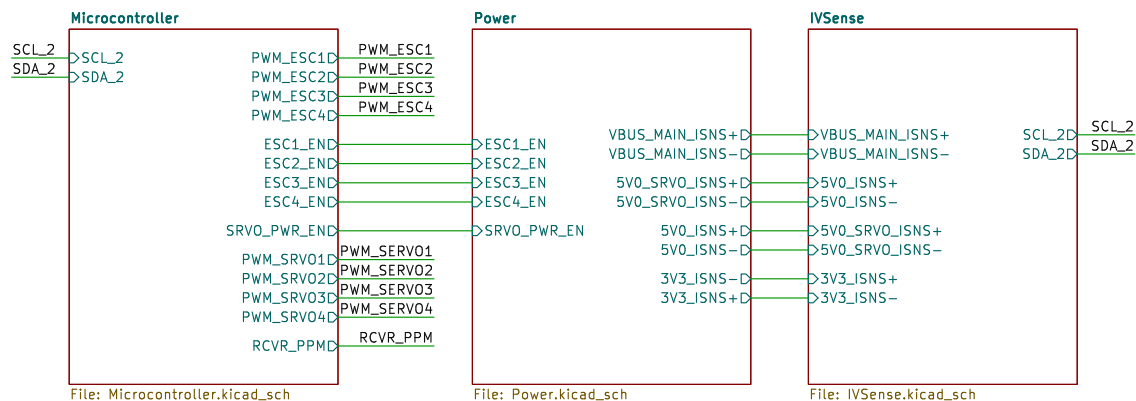


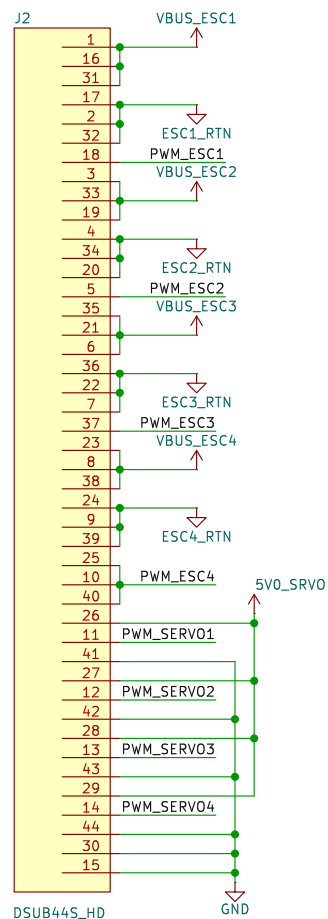
Info:

Requirements:

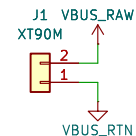
- Shall operate between 14V – 36V with nominal operation of 22V
- Shall contain reverse polarity protection
- Shall contain voltage transient/surge protection from 50V OV conditions
- Shall contain undervoltage lockout at 12V
- Shall contain soft start current protection
- Shall contain Inrush current limiter
- Shall leverage teensy 4.0 microcontroller daughter board
- Shall be able to operate four 9G servos with each 2A stall current
- Shall be able to operate have four ESC with 30A max each
- Target aircraft host shall be HEEWING F-01 Ultra Delta Wing 690mm EPP RC Airplane



Main Output



Batt Input



Mounting Holes



Sheet: /

File: Keres.kicad_sch

Title: Keres Flight Control Board Phase I

Size: A

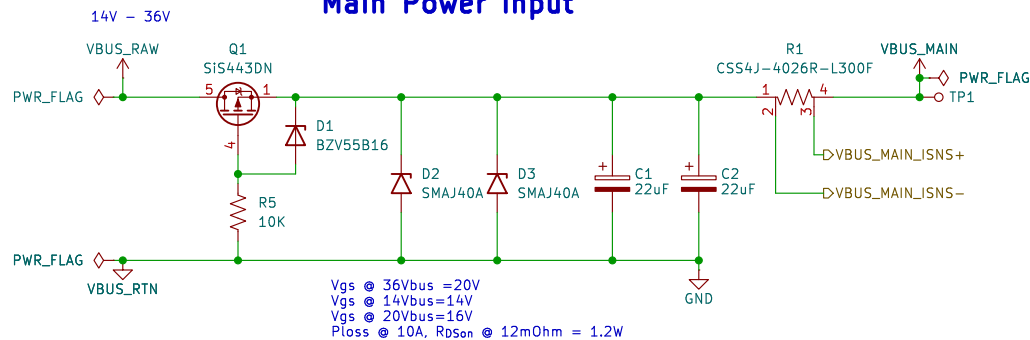
Date: 2026-04-06

Rev: A

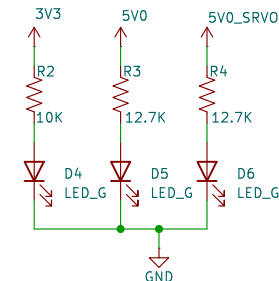
KiCad E.D.A. 9.0.7

Id: 1/4

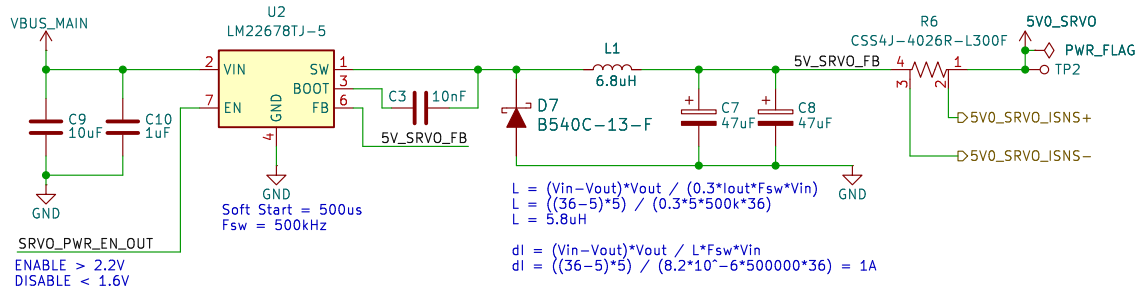
Main Power Input



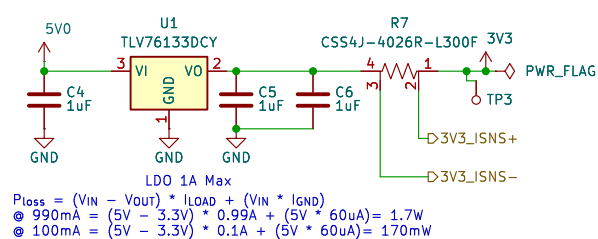
LED Health Indicators



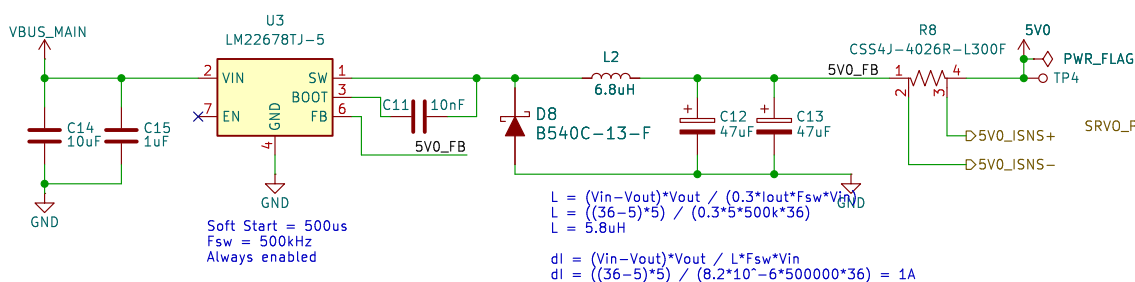
5V0 Servo Power Bus



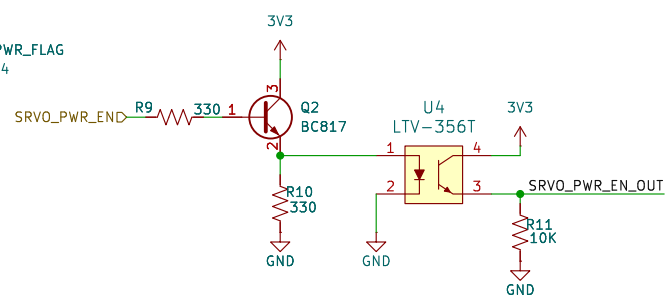
3V3 LDO Power



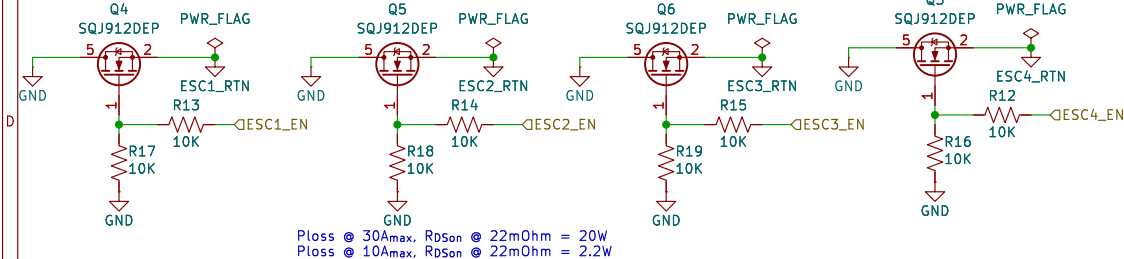
5V0 Power Bus



Servo Power Enable



ESC Power Enable



Sheet: /Power/
File: Power.kicad_sch

Title: Keres Flight Control Board Phase I

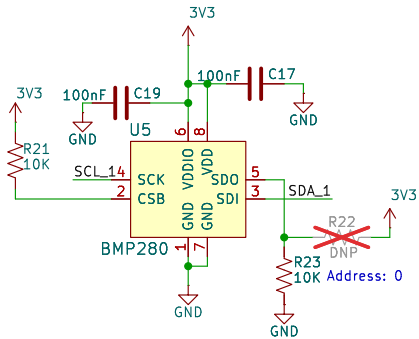
Size: A Date: 2026-04-06

KiCad E.D.A. 9.0.7

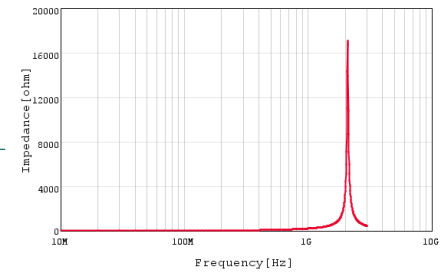
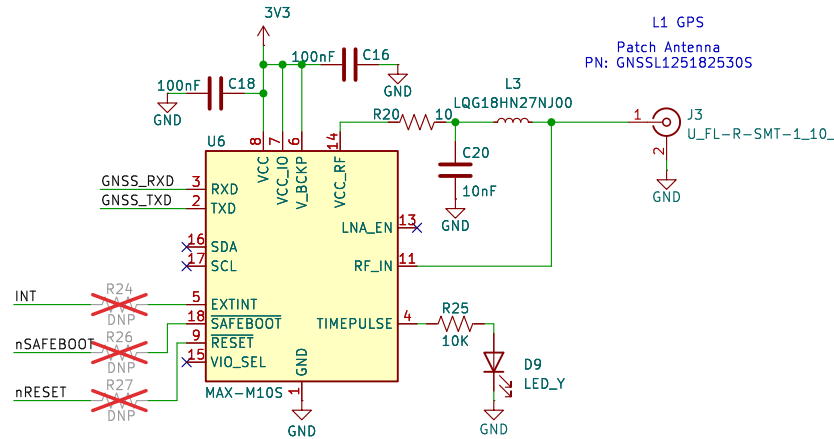
Rev: A

Id: 2/4

BMP280 – Pressure Sensor



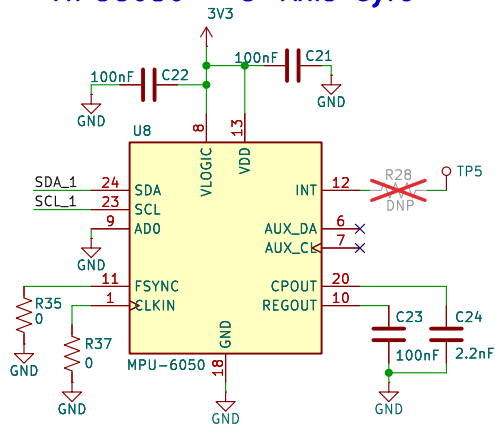
u-blox MAX-M10S GNSS



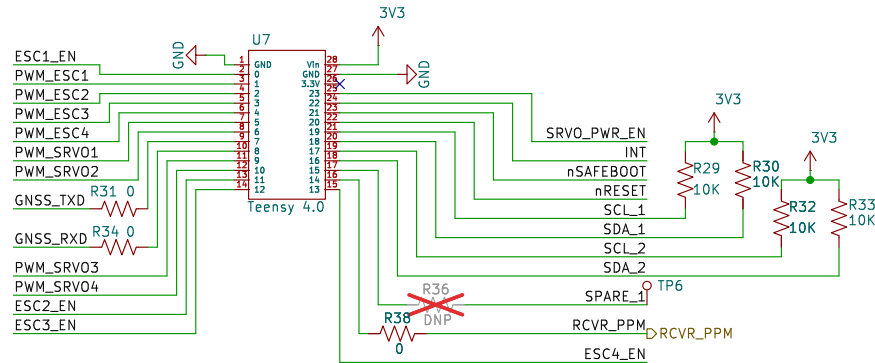
Sheet Pins

SRVO_PWR_EN	▷SRVO_PWR_EN
ESC1_EN	▷ESC1_EN
ESC2_EN	▷ESC2_EN
ESC3_EN	▷ESC3_EN
ESC4_EN	▷ESC4_EN
PWM_ESC1	▷PWM_ESC1
PWM_ESC2	▷PWM_ESC2
PWM_ESC3	▷PWM_ESC3
PWM_ESC4	▷PWM_ESC4
PWM_SRV01	▷PWM_SRV01
PWM_SRV02	▷PWM_SRV02
PWM_SRV03	▷PWM_SRV03
PWM_SRV04	▷PWM_SRV04
SCL_2D	▷SCL_2
SDA_2D	▷SDA_2

MPU6050 – 6-Axis Gyro



Teensy uC 4.0 Dev Board



Sheet: /Microcontroller/
File: Microcontroller.kicad_sch

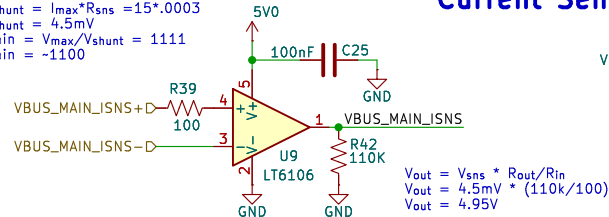
Title: Keres Flight Control Board Phase I

Size: A Date: 2026-04-06
KiCad E.D.A. 9.0.7

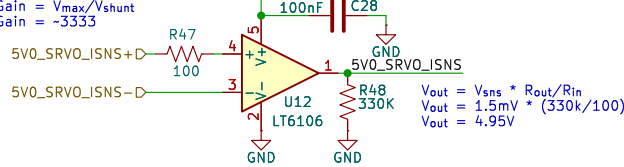
Rev: A
Id: 3/4

$I_{max} = 15A$
 $V_{shunt} = I_{max} \cdot R_{sns} = 15 \cdot 0.0003$
 $V_{shunt} = 4.5mV$
 $Gain = V_{max}/V_{shunt} = 1111$
 $Gain = -1100$

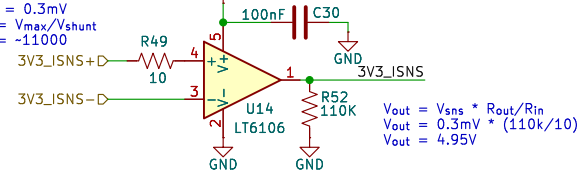
Current Sense



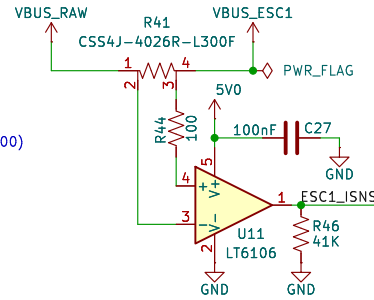
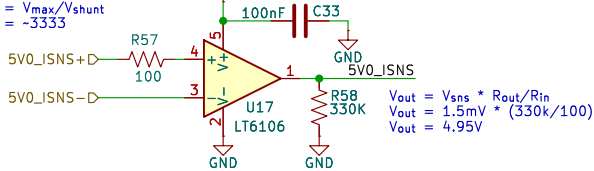
$I_{max} = 5A$
 $V_{shunt} = I_{max} \cdot R_{sns} = 5 \cdot 0.0003$
 $V_{shunt} = 1.5mV$
 $Gain = V_{max}/V_{shunt} = 333.3$
 $Gain = -333.3$



$I_{max} = 1A$
 $V_{shunt} = I_{max} \cdot R_{sns} = 1 \cdot 0.0003$
 $V_{shunt} = 0.3mV$
 $Gain = V_{max}/V_{shunt} = 1100$
 $Gain = -1100$

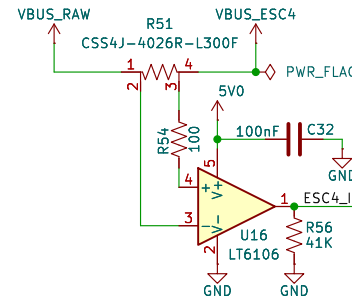
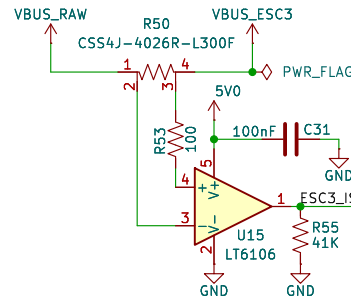


$I_{max} = 5A$
 $V_{shunt} = I_{max} \cdot R_{sns} = 5 \cdot 0.0003$
 $V_{shunt} = 1.5mV$
 $Gain = V_{max}/V_{shunt} = 333.3$
 $Gain = -333.3$

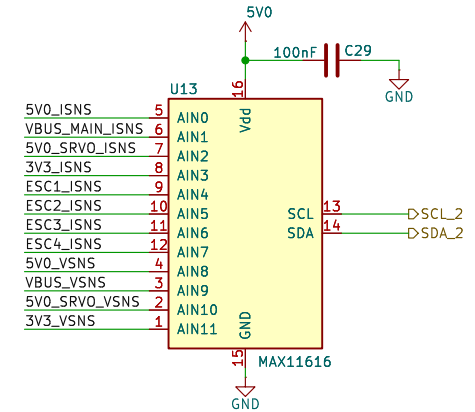


$I_{max} = 40A$
 $V_{shunt} = I_{max} \cdot R_{sns} = 40 \cdot 0.0003$
 $V_{shunt} = 12mV$
 $Gain = V_{max}/V_{shunt} = 417$
 $Gain = -410$

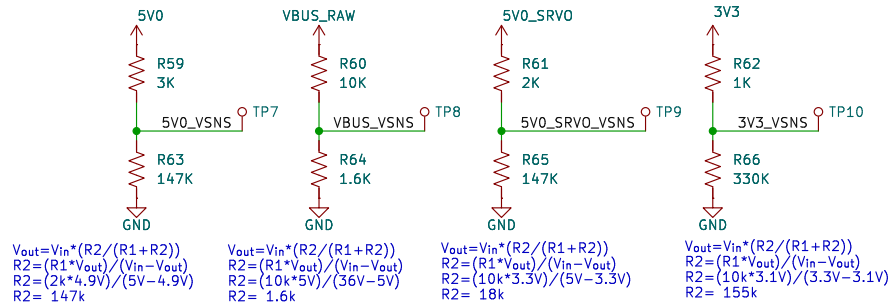
$V_{out} = V_{sns} \cdot \frac{R_{out}}{R_{in}}$
 $V_{out} = 112mV \cdot (41k/100)$
 $V_{out} = 4.92V$



Sense ADC



Voltage Sense



12-Bit ADC, 5V_{ref}
 5V/4095 = 1.22mV / Volt

Sheet: /IVSense/
 File: IVSense.kicad_sch

Title: Keres Flight Control Board Phase I

Size: A4 Date: 2026-04-06

KiCad E.D.A. 9.0.7

Rev: A

Id: 4/4